

GÖTEBORGS UNIVERSITET

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Blockchain in Trade Documentation

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Abstract

This paper explores the integration of blockchain technology into global trade documentation, particularly electronic bills of lading (eBLs), and evaluates its potential to modernize trade systems by addressing inefficiencies, fraud risks, and legal uncertainties inherent in traditional paper-based processes. Blockchain's decentralized, transparent, and immutable framework enhances the core functionalities of trade instruments, offering enhanced efficiency, security, and reliability. However, inconsistent legal frameworks, fragmented technical standards, and infrastructure limitations, especially in developing nations present significant barriers to its adoption.

Key legal frameworks, including the UNCITRAL Model Law on Electronic Transferable Records (MLETR) and the UK's Electronic Trade Documents Act 2023, provide pathways for recognizing digital trade documents, but global harmonization remains elusive. Case law analysis, including Mainline Private Hire Ltd v. Anthony Nolan and Glencore International AG v MSC Mediterranean Shipping Company SA, highlights the complexities of applying traditional concepts such as possession and control to blockchain systems. The study also examines platforms like Bolero and TradeLens, which illustrate the challenges of fragmented "digital islands" and the need for interoperable systems supported by universally accepted technical standards and governance frameworks.

Investments in infrastructure, capacity-building, and legal reform are essential to enable inclusive participation in digital trade. By fostering international collaboration, legal harmonization, and interoperable platforms, blockchain technology has the potential to revolutionize global trade systems, making them more efficient, secure, and equitable for all stakeholders.

Abstract **2**
Introduction **3**
Statement of Issues and Problems **4**
The Inefficiencies of Traditional Documentation Systems 4
Inconsistent Legal Recognition of Digital Trade Documents 4
Relevant Factual Elements **6**
Legal and Operational Challenges in Blockchain Integration 6
Blockchain in Trade **7**
Application of Law to Facts **9**
A Hypothetical Case 9
Case Law Analysis 10
Critique of Existing Works and Opinions **11**
Conclusion **12**
Bibliography **13**

Introduction

International trade has long been a cornerstone of economic development, enabling the exchange of goods and services across borders and fostering interconnectivity between nations. However, the infrastructure that supports international trade, particularly in the realm of trade documentation, remains deeply rooted in traditional paper-based systems. Instruments such as bills of lading, letters of credit, and customs declarations play critical roles as evidence of contracts, documents of title, and guarantees of payment. Despite their historical significance, these systems are increasingly viewed as impediments to efficiency in the global trading system. Lord Hoffmann described the bill of lading as "evidencing a contract of carriage but also a document of title".[1]

The inefficiencies of traditional trade documentation extend beyond operational delays. They also expose stakeholders to fraud, disputes, and legal uncertainties, particularly in the determination of possession and ownership. For instance, the Carriage of Goods by Sea Act 1971 in the UK and the Hague-Visby Rules govern the carriage of goods by sea, emphasizing the enforceability of bills of lading in ensuring delivery of goods to the rightful owner.[2] However, these laws were designed for a pre-digital age, necessitating significant adaptations to address the realities of modern trade.

Blockchain technology has emerged as a revolutionary solution to these challenges. The WTO observes, "digital trade fosters growth by improving resource allocation and reducing transaction costs".[3] Blockchain's decentralized, immutable, and transparent framework allows for secure verification and transfer of digital trade documents, eliminating the need for traditional intermediaries. By using blockchain, trade documentation can achieve the functional equivalence of paper-based systems, as recognized by the UNCITRAL Model Law on Electronic Transferable Records. The MLETR's principles of technology neutrality and functional equivalence provide a legal foundation for the recognition of digital transferable records, aligning legal frameworks with the realities of modern commerce.[4]

However, the legal and regulatory landscape remains fragmented. For example, the UK's Electronic Trade Documents Act 2023 aims to provide explicit recognition of digital trade documents, addressing gaps in the existing legal framework[5]. Similarly, Singapore has integrated the MLETR into its Electronic Transactions Act, offering a robust model for legal reform[6]. Despite these advancements, global harmonization remains a significant challenge, with inconsistent recognition of digital records across jurisdictions.

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[1] Owners of Cargo Lately Laden on Board the Ship or Vessel "Starsin" v Owners and/or Demise Charterers of the Ship or Vessel "Starsin" [2003] UKHL 12, [2001] EWCA Civ 56 (HL)
[2] Carriage of Goods by Sea Act 1971, c.19
[3] World Trade Organization, *Digital Trade for Development* (WTO 2023) accessed [4 January 2025]
[4] UNCITRAL, 'Model Law on Electronic Transferable Records 2017 accessed [2 January 2025]
[5] Electronic Trade Documents Act 2023, c.38
[6] Hannah Nguyen, 'Trade Digitization on a Global Scale: How Far Are We?' (World Customs Organization, 24 February 2022)) accessed [2 January 2025]

This essay explores how blockchain technology addresses the inefficiencies of traditional trade documentation and assesses the regulatory challenges that hinder its widespread adoption. By analyzing international legal frameworks, including the MLETR, and case studies such as TradeLens and Bolero, the paper aims to provide actionable recommendations for fostering a regulatory environment conducive to blockchain integration in global trade. Blockchain's transformative potential lies not just in its technological innovation but in its capacity to redefine legal and regulatory paradigms, paving the way for a more efficient, secure, and inclusive trading system.

Statement of Issues and Problems

The Inefficiencies of Traditional Documentation Systems

Trade documentation has historically been at the core of international commerce, performing critical roles such as verifying ownership, evidencing contracts, and enabling secure payment. As the bill of lading is described by Lord Hoffman, "evidencing a contract of carriage but also a document of title".[7]
Paper-based systems remain the dominant framework, introducing vulnerabilities such as delays in the verification of documents, risks of forgery, and high operational costs. 4 billion documents in constant circulation are facilitating trade today. Less than 1 percent of all trade documents are digital and the rest are in paperform. A single transaction requires 36 different documents in 240 copies that are distributed between more than 30 parties, on average.[8] To further highlight the inefficiencies of today's paper based system, transportation between Jakarta and Singapore takes one day, but the documentation process can take up to seven[9].

The MLETR aims to address these inefficiencies by establishing principles of technology neutrality and functional equivalence. It provides a legal framework for electronic transferable records to achieve the same validity as their paper counterparts, facilitating a shift toward digitized trade systems.[10]

Inconsistent Legal Recognition of Digital Trade Documents

The Model Law on Electronic Transferable Record enables the legal use of electronic transferable records by ensuring they are functionally equivalent to traditional paper-based instruments like bills of lading, promissory notes, and warehouse receipts. These records

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[7] Owners of Cargo Lately Laden on Board the Ship or Vessel "Starsin" v Owners and/or Demise Charterers of the Ship or Vessel "Starsin" [2003] UKHL 12, [2001] EWCA Civ 56 (HL)
[8] International Chamber of Commerce (ICC), *'Legal Status of Digitalisation of International Trade' 2024* accessed [2 January 2025]
[9] ICC Sweden, Henrik Blomqvist, *The Digitalisation of Global Trade* (School of Business, Economics and Law, Gothenburg University)
[10] United Nations Commission on International Trade Law (UNCITRAL), Model Law on Electronic Transferable Records (2017).

entitle the holder to claim performance of obligations and allow the transfer of claims through possession, fostering cross-border and domestic digital trade[11].

The Rotterdam Rules (2009) offer a complementary framework for modernizing trade documentation by explicitly recognizing electronic transport records. These rules require that electronic records meet standards of integrity and reliability equivalent to paper documents, supporting their use in international carriage.[12] However, the Rotterdam Rules face limited adoption globally, restricting their ability to harmonize the treatment of digital trade documents across jurisdictions[13]. Despite its potential, the adoption of the MLETR also remains limited. Currently, only a handful of jurisdictions, including Singapore through their Electronic Transactions Act (2010), Paraguay, Kiribati, Belize, Bahrain, and Abu Dhabi Global Market, have legally recognized electronic bills of lading by enacting the MLETR[14]. This implies that even when parties choose to utilize digital solutions for exchanging bills of lading, they frequently need to revert to physical copies upon reaching jurisdictions where electronic transferable records (ETRs) are not yet legally recognized. Such disparities in legal recognition hinder the global interoperability of digital trade systems, limiting the efficiency and security gains that electronic documentation promises.

In the United Kingdom, traditional trade documents like bills of lading are critical instruments in international commerce. Interpreted under the Carriage of Goods by Sea Act 1992, a bill of lading serves as a receipt for goods, evidence of the contract of carriage, and a document of title[15]. However, the legal enforceability of such documents traditionally relies on their tangible form, requiring physical endorsement and delivery for the transfer of ownership. New regulation in the United Kingdom, the Electronic Trade Documents Act (2023), aims to define an electronic trade document so as to achieve the same legal status as a paper-based trade document[16]. International conventions like the Hague-Visby Rules, incorporated into UK law through the Carriage of Goods by Sea Act 1971, govern key aspects of the obligations, rights, and liabilities of carriers and shippers in contracts of carriage evidenced by bills of lading. These rules emphasize the physical nature of trade documentation, requiring paper-based bills to function as legally binding instruments[17]. Similarly, the United Nations convention on the Carriage of Goods by Sea (1978), Hamburg Rules, which are recognized in many jurisdictions, offer an alternative regime for carrier liability and contracts of carriage by sea but also maintain a paper-centric approach[18]. These

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[11] United Nations Commission on International Trade Law (UNCITRAL), Model Law on Electronic Transferable Records 2017 accessed [4 January 2025]

[12] United Nations Convention on Contracts for the International Carriage of Goods Wholly or Partly by Sea (Rotterdam Rules) (adopted 11 December 2008, not yet in force)

[13] United Nations Commission on International Trade Law (UNCITRAL), Model Law on Electronic Transferable Records 2017

[14] Hannah Nguyen, 'Trade Digitization on a Global Scale: How Far Are We?' (World Customs Organization, 24 February 2022)) accessed [2 January 2025]

[15] Carriage of Goods by Sea Act 1992, c. 50, s 1(2)(a) and 2(1)(a)

[16] Electronic Trade Documents Act 2023, c.38

[17] Carriage of Goods by Sea Act 1971, c.19

[18] United Nations Convention on the Carriage of Goods by Sea (Hamburg Rules) 1978

frameworks, while fundamental to ensuring consistency and fairness in maritime trade, were established in a pre-digital era and do not address the complexities of electronic records or blockchain-based documentation systems.

Relevant Factual Elements

Legal and Operational Challenges in Blockchain Integration

The lack of coordination in digitization efforts has led to the emergence of multiple blockchain-based platforms, such as TradeLens, and Bolero, each operating as isolated "digital islands" with its own set of technical standards and protocols for data digitization and exchange. This fragmented ecosystem makes it possible to establish one-to-one connections between parties, but scaling to one-to-many connections becomes prohibitively costly and complex, especially in the context of cross-border trade. For instance, the electronic bill of lading, despite the first system being introduced as early as 1998, more than a dozen eBL platforms now exist, each functioning independently. In practice, all parties involved in a single transaction must subscribe to the same platform for seamless interoperability[19]. This arrangement imposes a significant operational burden on companies engaged in diverse trade relationships, as they would need to subscribe to multiple platforms to accommodate different counterparties. Such a setup is neither cost-effective nor practical for most businesses, particularly small and medium-sized enterprises (SMEs). Efforts like the ICC Digital Standards Initiative (DSI) aim to address this fragmentation by promoting globally accepted standards for digital trade documentation and blockchain integration[20]. However, until such standards are universally adopted, the lack of interoperability among platforms remains a major barrier to realizing the full potential of blockchain in global trade.

The legal recognition of blockchain-based records faces significant challenges, particularly in jurisdictions that have not adopted frameworks like the UNCITRAL Model Law on Electronic Transferable Records. Courts in these jurisdictions will lack established precedent to address disputes involving blockchain records, creating uncertainty about their legal enforceability. This gap in precedent raises critical questions about the validity, ownership, and transferability of blockchain-enabled trade documents in cross-border transactions. Adding to these complexities, many developing nations face significant hurdles, as they lack the robust legal frameworks necessary to enact reforms that would legally recognize ETRs[21].Furthermore, broader barriers to adopting digital technologies compound the issue, including inadequate infrastructure such as unreliable electricity, limited internet connectivity, and underdeveloped telecommunications networks. Restricted access to affordable devices and internet services also inhibits the widespread use of digital systems.

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[19] Hannah Nguyen, 'Trade Digitization on a Global Scale: How Far Are We?' (World Customs Organization, 24 February 2022)) accessed [2 January 2025]
[20] ICC Digital Standards Initiative, 'Our Work' accessed [3 January 2025]
[21] Hannah Nguyen, 'Trade Digitization on a Global Scale: How Far Are We?' (World Customs Organization, 24 February 2022)) accessed [2 January 2025]

Additionally, a lack of digital literacy and the skills required to navigate these technologies effectively further impede efforts to modernize trade systems in these regions[22].

In many jurisdictions, traditional laws governing trade documentation such as the Bills of Exchange Act 1882 in the UK do not account for the unique characteristics of blockchain technology and the concept of digital possession[23]. Unlike paper-based documents, which require physical custody for possession to be legally recognized, blockchain records rely on cryptographic keys to grant access and control. The principles of "factual possession" and "intention to possess," as articulated in Mainline Private Hire Ltd v. Anthony Nolan, highlights the complexity of applying traditional notions of possession to digital records. In that case, the court emphasized that possession requires both physical custody and an intention to exercise control for one's benefit[24]. Blockchain systems, which lack physical custody, challenge this legal framework by redefining possession in terms of digital control and cryptographic authorization. Moreover, the decentralized nature of blockchain complicates the exclusivity requirement. For example, in Mainline, the court found that exclusive possession was crucial to determining ownership and control of a physical asset. Translating this principle to blockchain raises questions about whether possession is held by the private key holder, the platform operator, or the network participants as a collective. This creates uncertainty about whether blockchain records can fulfill the legal standards for possession and transferability in trade.

Even in jurisdictions that have adopted the MLETR, courts may be reluctant to establish binding precedent due to the novelty of blockchain technology. Judges may lack technical expertise or rely on traditional legal principles, delaying the development of robust case law supporting blockchain's use in trade. This reluctance leaves businesses uncertain about the legal risks associated with adopting blockchain solutions.

Blockchain in Trade

Blockchain is a distributed ledger technology that records transactions across a decentralized network of computers in a secure, transparent, and immutable manner. Each block in the chain contains a list of transactions, and these blocks are linked cryptographically, ensuring that no block can be altered without consensus from the network participants. This decentralized nature eliminates the need for intermediaries, as transactions are validated collectively by the network. Blockchain's use of cryptographic keys ensures secure and exclusive control over data. These features make blockchain particularly suitable for managing digital assets, ETRs such as eBLs. In the context of trade, blockchain's decentralized structure offers the ability to replicate key functionalities of traditional paper-based systems, including evidencing contracts, serving as documents of title, and enabling secure and efficient transfer of ownership.

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[22] World Trade Organization, *Digital Trade for Development* (WTO 2023) accessed [4 January 2025]
[23] Bills of Exchange Act 1882, c. 61
[24] Mainline Private Hire Ltd v Nolan [2011] EWCA Civ 189, Case No: B2/2009/1195, 2011 WL 664524 (CA)

Despite the transformative potential of blockchain, most digital trade systems today remain centralized.[25] Centralized systems rely on a single governing entity to oversee the issuance, transfer, and management of ETRs. These systems often function within a closed network where participants must agree to specific contractual frameworks to engage in transactions.[26] While centralized platforms offer structure and legal certainty, they also introduce limitations, such as reliance on a single point of control and challenges in interoperability with decentralized solutions like blockchain.

Bolero operates as a centralized system for managing electronic bills of lading (eBLs), where a central registry oversees their issuance and transfer. The system relies on mutual trust among its members, who must join by agreeing to a multipartite contract framework known as the Bolero Rulebook.[27] This Rulebook creates a contractual relationship among the users, addressing issues related to the privity of contracts and enabling third parties to acquire rights linked to the eBL. Under the Bolero system, the Rulebook explicitly recognizes eBLs as equivalent to traditional paper-based bills of lading (PBLs)[28]. These electronic documents serve as receipts for goods, evidence of the contract of carriage, and documents of title that must be presented to claim delivery of goods. One of Bolero's key features is its process for transferring rights, which is facilitated through "novation." This process involves substituting the contractual obligations under the original carriage agreement with new terms, enabling the eBL to replicate the functionality of a PBL.[29]

Unlike decentralized blockchain systems, Bolero requires all transactions to pass through its central title registry, which ensures compliance with the Rulebook and maintains the integrity of the system.[30] While this centralized design ensures structure and control, it also limits the system's flexibility and interoperability with other platforms. Bolero remains a significant example of how centralized digital trade solutions attempt to modernize traditional practices while adhering to established legal standards.

TradeLens, developed by Maersk and IBM, aimed to modernize global trade by providing a blockchain-enabled platform for managing shipping documentation and supply chain data. By using blockchain, it sought to enhance transparency, efficiency, and security in international trade. However, TradeLens operated as a partially centralized system, with Maersk serving as both a platform provider and participant, raising concerns over conflicts of interest and data control[31].

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[25] Tianyi Jiang, *Regulating Electronic Bills of Lading on a National Level* (2023) 7 JBL 604-606
[26] Tianyi Jiang, *Regulating Electronic Bills of Lading on a National Level* (2023) 7 JBL 604-606
[27] Bolero Rulebook art.2.1.2(1)
[28] Tianyi Jiang, *Regulating Electronic Bills of Lading on a National Level* (2023) 7 JBL 604-606
[29] Bolero Rulebook arts 3.1(1), 3.6(1)
[30] Bolero Rulebook art.1.1(53)
[31] Ilias Ioannou, *Relational Trade Networks* (2023) 82(3) C.L.J., 450-482

Despite its promise, TradeLens faced significant challenges. The "co-opetition paradox" (cooperation with competitors) deterred competitors from joining due to fears of losing control over sensitive data. Additionally, the platform's rulebook allowed unilateral modifications to terms, increasing uncertainty for users.High integration costs and the absence of harmonized international frameworks for blockchain-based systems further hindered adoption[32].

TradeLens's failure in 2022 highlights the limitations of blockchain in multi-stakeholder environments without robust legal frameworks and governance structures. Its challenges emphasize the need for balanced systems that address trust, fairness, and regulatory alignment for blockchain-based trade solutions to succeed.

Application of Law to Facts

A Hypothetical Case

A UK-based exporter ships machinery to a buyer in Sweden. Both parties agree to use an electronic bill of lading stored on a blockchain platform compliant with the UK's Electronic Trade Documents Act 2023[33]. Upon arrival in Sweden, Swedish customs authorities reject the eBL, citing the lack of explicit legal recognition for electronic transferable records under Swedish law. The buyer claims non-delivery due to customs' rejection and refuses to release payment, arguing that the blockchain-based eBL is not a legally valid document of title. The exporter counters that ownership was transferred via the blockchain upon shipment, as per the agreed terms.

The United Kingdom's Legal Framework provides a progressive foundation for recognizing electronic trade documents. The Electronic Trade Documents Act 2023 stipulates that electronic trade documents, such as electronic bills of lading, must meet specific criteria to achieve legal equivalence with their paper counterparts. These criteria include reliability, integrity, and exclusive control, which ensure that the electronic document functions effectively as a document of title[34]. While the Act does not explicitly address blockchain-based systems, the exporter could argue that a well-designed blockchain system satisfies these requirements. For instance, the immutable nature of blockchain can demonstrate reliability and integrity, while cryptographic keys can establish exclusive control, fulfilling the Act's requirements for a valid document of title under UK law.
The Carriage of Goods by Sea Act 1992 further enhances this framework by confirming the use of bills of lading, to evidence contracts of carriage and facilitate ownership transfer upon shipment[35]. Blockchain systems that employ secure cryptographic keys to manage access and control could align with COGSA's principles, enabling integration of eBLs into international trade transactions.

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[32] Ilias Ioannou, *Relational Trade Networks* (2023) 82(3) C.L.J., 450-482
[33] Electronic Trade Documents Act 2023, c. 38, s2(2)
[34] Electronic Trade Documents Act 2023, c. 38, s2(1) and 5(1)
[35] Carriage of Goods by Sea Act 1992, c. 50, s 2(1), 5(1).

The Swedish legal framework poses significant challenges for blockchain-based trade. The Swedish Maritime Act (Sjölag 1994:1009), incorporating the Hague-Visby Rules, remains paper-centric and does not explicitly recognize electronic transferable records[36]. Similarly, while the eIDAS Regulation governs electronic signatures, it does not extend to comprehensive recognition of eBLs or blockchain-based documentation, creating uncertainty in cross-border transactions involving Sweden[37].

The case of Glencore International AG v MSC Mediterranean Shipping Company SA further illustrates the complexities of integrating digital systems into traditional trade practices. It addressed whether electronic release codes fulfilled delivery obligations under a bill of lading[38], emphasizing the importance of clear contractual provisions to validate electronic processes. Similar disputes could arise in the blockchain-based eBL scenario, where the absence of explicit agreements or statutory recognition, such as in Sweden, could compromise enforceability. This case highlights the need for robust legal frameworks and explicit contractual terms to bridge the gaps in jurisdictions transitioning to digital trade systems to harmonize and integrate it for world trade.

Case Law Analysis

The treatment of electronic trade documentation in courts has varied, highlighting the legal complexities in adapting traditional frameworks to modern digital systems. In a landmark Swedish case from 2017 (Högsta Domstolen, Ö 5072-16), the Supreme Court addressed the validity of electronic promissory notes as transferable records. The court affirmed that electronic records could potentially satisfy the requirements of promissory note law if they included protective mechanisms equivalent to those provided by possession and original documentation in paper-based systems. However, it concluded that existing technical solutions were inadequate.[39] This decision underscored the need for stronger technological and legal alignment to accommodate electronic transferable records.

A more recent case in 2023 (Svea Court of Appeal, ÖÄ 8442-22) revisited this issue when a creditor sought enforcement of a claim based on an electronic promissory note. Although the court did not challenge the technical security of the solution, it rejected the note as a transferable record because it required an internet connection, which conflicted with the principle of technology neutrality.[40] The decision highlighted the challenges in reconciling legal standards with evolving technological capabilities, particularly the lack of clear criteria for electronic transferable records.

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[36] Sjölag (1994:1009)

[37] European Parliament and Council Regulation (EU) No 910/2014 of 23 July 2014 on electronic identification and trust services for electronic transactions in the internal market [2014] OJ L257/73

[38] Glencore International AG v MSC Mediterranean Shipping Co SA [2015] EWHC 1989 (Comm), Case No: 2013 FOLIO 424, 2015 WL 4172853 (QB)

[39] Swedish Enforcement Authority (Kronofogden), 2017. Supreme Court Ruling of 2 November 2017 on Electronic Promissory Notes, Ö 5072-16, Memorandum (in Swedish)

[40] Svea Court of Appeal, ÖÄ 8442-22, 23 February 2023, as referenced in *Digitalisation of International Trade: The Legal Status of Electronic Transferable Records in Sweden* (International Chamber of Commerce, August 2024)

In the English case of Glencore International AG v MSC Mediterranean Shipping Company SA, the court addressed whether an electronic release system fulfilled delivery obligations under a bill of lading. The ruling emphasized the necessity of explicit contractual provisions to validate electronic processes, especially regarding delivery mechanisms and liabilities. It also highlighted the risks of relying on digital systems without robust agreements, particularly in cross-border transactions where legal interpretations may differ[41]. The principles of "factual possession" and "intention to possess," as articulated in Mainline Private Hire Ltd v. Anthony Nolan, add another layer of complexity when applied to blockchain-based trade documentation. The court in this case emphasized that possession requires both physical custody and an intention to exercise control for one's benefit.[42] Blockchain systems, which lack physical custody, challenge this framework by redefining possession in terms of digital control and cryptographic authorization. Moreover, the decentralized nature of blockchain complicates the exclusivity requirement, as seen in Mainline, where the court found that exclusive possession was crucial to determining ownership and control of a physical asset[43]. Translating these principles to blockchain raises fundamental questions about whether digital systems can satisfy traditional legal standards for possession and transferability,

These cases collectively illustrate that while courts recognize the potential of electronic documentation, the lack of clear standards, reliance on technology-neutral principles, and insufficient legal harmonization continue to pose significant barriers. Courts demand that electronic systems achieve functional equivalence to traditional documents in areas like possession, control, and original documentation. These rulings highlight the importance of explicit agreements and global frameworks, such as the UNCITRAL Model Law on Electronic Transferable Records, in enabling the integration of digital solutions into international trade while preserving the legal protections of traditional frameworks.

Critique of Existing Works and Opinions

Nguyen's 2022 article, *Trade Digitization on a Global Scale: How Far Are We?*, Ioannou 2023, *Relational trade networks*, and Jiang 2023*, Regulating electronic bills of lading on a national level* provide valuable insights into the challenges and opportunities of blockchain-based trade documentation, though significant gaps persist. Nguyen highlights the fragmentation caused by "digital islands" of independent eBL platforms, which limit interoperability and impose burdens on SMEs. Her emphasis on harmonization through initiatives like the ICC Digital Standards Initiative is critical but under-explores the economic and infrastructural barriers faced by developing nations. Jiang complements this by critiquing

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[41] Glencore International AG v MSC Mediterranean Shipping Co SA [2015] EWHC 1989 (Comm), Case No: 2013 FOLIO 424, 2015 WL 4172853 (QB)

[42] Mainline Private Hire Ltd v Nolan [2011] EWCA Civ 189, Case No: B2/2009/1195, 2011 WL 664524 (CA)

[43] Mainline Private Hire Ltd v Nolan [2011] EWCA Civ 189, Case No: B2/2009/1195, 2011 WL 664524 (CA)

the slow adaptation of legal frameworks to electronic bills of lading (eBLs). He highlights the need for enforceable legal frameworks and points out that both centralized platforms like Bolero and decentralized approaches suffer from insufficient legal certainty. His focus on the legal dimensions aligns with Nguyen's emphasis on standardization but stresses the importance of scalability through robust enforcement mechanisms.

Ioannou adds another layer by examining governance challenges in blockchain trade platforms, highlighting the co-opetition paradox, where platform providers may exploit their control over users despite the promise of decentralization. While his insights are crucial, the focus on English law limits the broader applicability of his arguments. Together, these works paint a nuanced picture of the barriers to blockchain-based trade systems, emphasizing the need for harmonized standards, enforceable legal frameworks, and governance mechanisms that ensure equitable participation. Addressing these challenges collectively is essential to unlocking the full potential of blockchain in global trade.

Conclusion

This essay examines the potential of blockchain technology to improve trade documentation by addressing inefficiencies such as delays, fraud risks, and legal uncertainties. However, its integration into global legal frameworks is incomplete, with fragmented laws and limited adoption of international standards creating barriers to broader use.

A significant issue is the lack of harmonized global standards. Frameworks like the UNCITRAL Model Law on Electronic Transferable Records establish functional equivalence and technology neutrality for electronic transferable records. However, adoption of the MLETR has been limited, with many jurisdictions, including Sweden, yet to implement its principles. Countries should prioritize integrating the MLETR into domestic laws and aligning it with existing trade regulations. Importantly, current reforms do not specifically define blockchain technology as a form of ETR, instead treating it as equivalent to paper-based systems, which limits clarity and creates gaps in application.

Legal analysis highlights the complexities of applying traditional concepts like possession to digital systems. Cases such as Mainline Private Hire Ltd v. Anthony Nolan demonstrate how blockchain challenges these principles by redefining possession in terms of digital control and cryptographic authorization. Policymakers need to address these issues by clarifying how blockchain can meet legal standards for possession and transferability.

To foster global adoption, interoperability and governance issues must also be resolved. Initiatives like the ICC Digital Standards Initiative should work toward creating unified platforms that ensure consistency across jurisdictions. At the same time, infrastructure gaps in developing nations must be addressed through targeted investments in technology and legal reforms to support digital trade.

Blockchain holds transformative potential for modernizing trade documentation, but realizing its potential requires coordinated legal, technical, and infrastructural reforms. Harmonizing standards and addressing disparities in adoption will be critical to building a robust and inclusive digital trade system.

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1.1 Task Distribution, Strengths, and Difficulties of the Essay Writing Process

Nicholas Tilk
I focused on the legal implications of blockchain in trade, examining how platforms like TradeLens and Bolero align with international legal frameworks.

Mauritz Andersson
I concentrated on analyzing legal frameworks such as UNCITRAL MLETR and the UK's Electronic Trade Documents Act 2023, alongside relevant case law. I focused on applying international trade law to blockchain's regulatory challenges.

1.2 Individual Contribution and Reflection

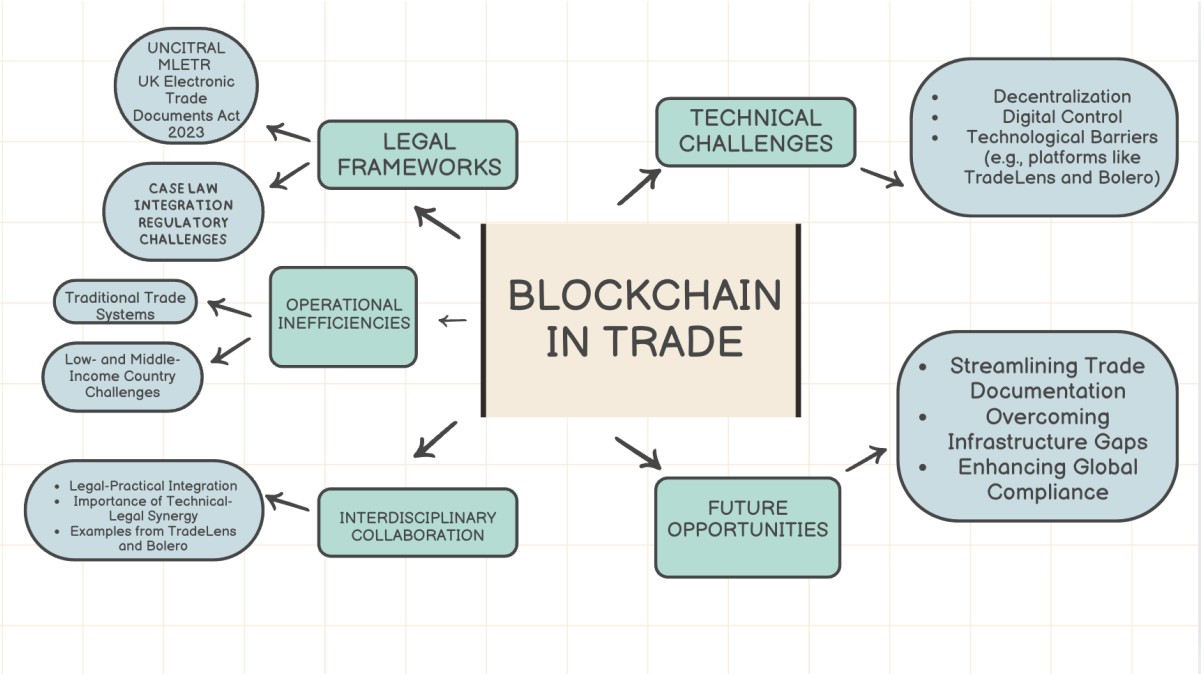
Nicholas Tilk
My primary focus was on evaluating digital platforms like Bolero and TradeLens and critiquing opinions of others. We both looked for relevant sources and cases relevant to our essay.

Mauritz Andersson
I concentrated on analyzing international legal frameworks like UNCITRAL MLETR, the UK's Electronic Trade Documents Act 2023, and relevant case law. We both discussed legal challenges, focusing on issues like possession and title.

Joint Reflection
Our collaboration was key to the essay's success. We shared tasks like brainstorming, drafting, and editing, combining our expertise to balance legal, technical, and practical perspectives.

1.3 preparatory document

2.0 Mindmap



3.0 AI Declaration

We used AI tools collaboratively during this essay to improve clarity. We used AI to enhance the structure and readability of our writing to make the essay cohesive and well-presented. However, all research, analysis, and critical thinking were conducted by us, with AI serving as a supplementary tool to refine and polish the final output.